Voting with Correlated Information

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Abstract

We study information aggregation in elections when voters' signals are correlated. We model exchangeable binary signals and heterogeneous voter biases, allowing the common signal quality to be arbitrarily close to uninformative with positive probability (the "bad-information case"). In this setting, correlation has sharp strategic implications: conditional on being pivotal in a large electorate, a voter infers that a close election is most likely to arise when signals are nearly uninformative. As a result, even an arbitrarily small voter bias makes informative voting fail. We then test the key comparative statics in a large online experiment that implements the correlated signal structure and varies committee size and, orthogonally, elicit contingent choices that make pivotality salient (via a strategy method treatment). While efficiency is high, especially in large groups, individual voting patterns do not display the pivotal scepticism central to the theory. Instead, behaviour is consistent with voters largely neglecting correlation.

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1 Introduction

Information correlation is a fact of electoral life. Voters often draw on the same newspapers, TV channels, websites, party cues, and social media feeds, and those outlets themselves rely on a small set of news agencies. Correlation matters for how a close election should be interpreted: when many voters receive (essentially) the same information, a tight vote is not just a large sample of independent signals. The contribution of this paper is to show that correlation can have sharp equilibrium effects and to examine whether the reasoning that generates them is behaviourally operative.

We study elections in which voters choose between two candidates and the most appropriate candidate is unknown. Each voter receives a private signal about the state, but signals are correlated across voters conditional on the state. Voters also have idiosyncratic preferences: they value selecting the correct candidate, but they also have a (possibly small) bias in favour of one candidate.

The key theoretical ingredient is that with some (possibly small) probability, the information environment is such that signals are essentially uninformative. Under this condition, we show that informative voting fails in large elections whenever voters have any non-zero bias, however small. The mechanism is driven by pivotal reasoning. In majority elections, a rational voter conditions on the event of being pivotal. If others vote informatively, pivotality requires that the population vote is essentially split 50/50. With correlated signals, that split is exponentially more likely when the underlying signal quality is near 1/2 than when signals are informative, even if the prior probability of uninformative signals is very small. Conditional on being pivotal, a voter therefore comes to believe that their own signal (and everyone else's) is almost uninformative, and so even an arbitrarily small bias becomes decisive. Neither correlation alone nor bias alone is sufficient for this collapse; it is the interaction of correlation, the possibility of uninformative signals, and pivotal reasoning.

To test this theoretical finding, we run a large online experiment ($N = 1123$).

Participants vote between two candidates (Orange and Lemon) and earn a payoff for choosing the candidate that matches the state, plus or minus a personal parameter that captures bias. Signals are correlated through a common shock to informativeness: with 10% probability all signals are uninformative (an independent fair coin), and with 90% probability all signals are informative with accuracy $\hat{q} \in \{0.8, 0.9, 0.95\}$ depending on the round. To understand that a close election signals a breakdown of informativeness and to act on this, participants need to (i) engage in pivotal reasoning and (ii) understand that pivotality implies a likely absence of information whenever group size is "large". To gain an insight into how these channels impact behaviour we vary group size ($N + 1 \in \{3, 15\}$) and, orthogonally, whether we use a strategy method that makes pivotality salient by asking participants to choose contingent on how others voted.

The experimental results are difficult to reconcile with the theoretical results. First, efficiency is high throughout, albeit with some variation. On average groups select the correct candidate more than 80% of the time. Second, at the individual level, many participants vote with their signal, and they do so especially when the signal points in the direction they already prefer. Third, making pivotality explicit through the strategy method has only modest effects: it does not raise efficiency and it does not push behaviour towards the sharp pivot-based logic behind the theoretical failure result. In particular, in the strategy-method data the response functions do not display the asymmetry around pivotality that our mechanism would imply. Participants react to others' votes in a smooth way and appear to treat additional votes as informative even right around the pivotal event. This is exactly the region where, under correlated information with possible uninformative signals, a pivotal voter should become sceptical of the informational content of the electorate's split.

Taken together, these results show that even if pivotal reasoning is made explicit and even if group size is substantial, participants do not react as predicted by theory. One possible interpretation is that participants simply neglect correlation all together. Under such reasoning, a close election does not primarily signal a breakdown of informativeness; instead it signals that the state is uncertain, so marginal pieces of information remain valuable. That logic predicts continued reliance on signals, not a retreat to bias, and it predicts response functions that look much closer to standard independent-signal models than to our correlated-information pivotality benchmark.

On the one hand, correlation and common sources of information are ubiquitous,

and the theory shows that even a small chance of uninformative signals can, in principle, make correlation decisive for equilibrium behaviour and information aggregation. On the other hand, if voters do not internalise correlation when they interpret close elections, then the most severe theoretical implications may be muted in practice. From this perspective, our paper delivers both a benchmark and a cautionary note: correlation is a powerful force in strategic models of voting, but it depends on voters' ability to understand pivotality and correlation, as well as to have a sense of the law of large numbers. In the experiment we vary group size and make pivotality explicit, but find that neither helps induce the pivotal skepticism predicted by the theory.

The rest of the paper proceeds as follows. Section 2 reviews the literature. Section 3 presents the model and theoretical results and derives the testable predictions used in the experimental design. Section 4 describes the experimental design and procedures, and Section 5 presents the experimental results. Section 6 discusses implications for voting theory with correlated information and for behavioural approaches to aggregation in elections.

2 Literature

Our theoretical analysis belongs to the literature on strategic information aggregation in elections. Foundational contributions establish how conditioning on pivotality can make informative voting inconsistent with equilibrium and characterise when large elections nevertheless aggregate dispersed information (Austen-Smith & Banks 1996, Feddersen & Pesendorfer 1996, 1997, Wit 1998, Myerson 1998, McLennan 1998, Duggan & Martinelli 2001). Work allowing heterogeneous preferences further identifies conditions for full-information equivalence and for failures of aggregation (Bhattacharya 2013, Barelli et al. 2022, Kosterina 2023). Our model adds exchangeable, conditionally correlated signals and private voter biases. Related experiments study strategic information aggregation and heterogeneous preferences (Mengel & Rivas 2017, Morton et al. 2019). Recent work also shows how privately correlated information supplied by a persuader can affect pivotal inference and electoral outcomes (Heese & Lauermann 2025).

The broader jury-theorem literature studies dependence among votes and signals under both sincere and strategic behaviour. Ladha (1992), Ladha (1993), Berg (1993), and Ladha (1995) show how correlation changes the information-aggregation prop-

erties of majority rule, while Peleg & Zamir (2012) and Pivato (2017) provide more general results for dependent juries and correlated voters. These contributions help isolate our mechanism: correlation by itself need not eliminate aggregation; here, the failure arises from the interaction of correlation, possible near-uninformative environments, private bias, and strategic conditioning on pivotality.

The closest theoretical paper is Mandler (2012), who also uses exchangeability and obtains failures of aggregation. Mandler imposes stronger restrictions on the behaviour of the de Finetti mixing distribution near its lower bound. Our result requires only positive probability in every right-neighbourhood of 1/2. A recent working paper by Odora (2024) also obtains asymptotically partisan voting with heterogeneous preferences, but through uncertainty about whether voters' interests are aligned rather than through correlated signal quality.

Our experiment connects to the literature on correlation neglect: the tendency to treat correlated pieces of information as independent, or at least to insufficiently discount redundancy. Early contributions include DeMarzo et al. (2003), Glaeser & Sunstein (2009), and Ortoleva & Snowberg (2015). Experimental and empirical evidence is provided by Eyster & Weizsäcker (2016), Enke & Zimmermann (2019), Grimm & Mengel (2020), Moser & Wallmeier (2021), and Hossain & Okui (2024). In the voting context, Levy & Razin (2015) show that correlation neglect can sometimes improve electoral outcomes. In our setting, by contrast, information aggregates without correlation when biases are sufficiently small, whereas even a small probability of near-uninformative common signal quality undermines informative voting if voters correctly condition on pivotality. Unlike this literature our experiment does not directly manipulate correlation, but instead is focused on understanding whether failure to reduce pivotal events and/or a lack of intuitive understanding of the law of large numbers - two key ingredients of the theoretical mechanism - contribute to correlation neglect in this particular setting.

Finally, the strategy-method treatments relate to experimental work on contingent and hypothetical reasoning. Esponda & Vespa (2014) distinguish difficulty with hypothetical thinking from difficulty extracting information from hypothetical events. Martínez-Marquina et al. (2019), Moser (2019), and Niederle & Vespa (2023) document broader failures of contingent reasoning. Our design applies these ideas to pivotal inference when signal quality is correlated across voters. There is not much prior research on how pivotal reasoning impacts voting decisions. Gerber et al. (2020)

find that voters update their beliefs about electoral competitiveness in response to polling information, but that these changes do not significantly affect voter turnout. This contrasts findings from Duffy & Tavits (2008) who in a lab experiment find that beliefs about pivotality do impact participation.

3 Theory

There is a pool of infinitely many potential voters, with each voter i characterised by a bias $b_i \in \mathbb{R}$, which is private information and drawn independently of the state and signal-generating process from an i.i.d. distribution G with full support on some interval $[a, b]$ with $a < 0 < b$. An electorate of $N+1 > 2$ voters, with N even, is formed by independent draws from this population; these voters have to decide between two candidates $c \in \{0, 1\}$ by simultaneously casting a vote for either candidate. The candidate that receives the most votes wins the election. Since $N + 1$ is odd, the final vote cannot be tied; the tie among the other N voters determines pivotality.

There is a state of nature that can take two values $s \in \{0, 1\}$, both equally likely a priori. The utility of each voter i when candidate c wins the election and the state is s is given by

$$u_i(c, s) \quad = \quad cs + (1 - c)(1 - s) + (2c - 1)b_i.$$

Thus, the voter receives one unit of utility whenever the winner of the election coincides with the state of nature plus a bias-dependent payoff. A voter gains $|b_i|$ from electing their preferred candidate and loses $|b_i|$ if the other candidate wins. Negative biases favour candidate 0 whereas positive biases favour candidate 1.

Before the election, each voter i receives a signal $\theta_i \in \{0, 1\}$. Each voter observes their own signal but not the common quality of the information environment. Conditional on the state, signals received by different voters may be correlated, and their joint distribution is common knowledge. A voter does not observe the state, the signals received by other voters, or the common signal quality.

We assume that, conditional on the state, the signals received by voters are anonymous, i.e., they do not depend on the identity of the voter. This implies that the sequence of signals is *exchangeable* (Johnson 1924):

Definition 1 (Exchangeability)**.** *The sequence of signals $\{\theta_i\}_{i=1}^{\infty}$ is exchangeable conditional on the state if, for every finite m, every permutation σ of $\{1, \ldots, m\}$, every*

vector $x \in \{0,1\}^m$*, and each state* $s \in \{0,1\}$*,*

$$P(\theta_1 = x_1, \ldots, \theta_m = x_m \mid s) \quad = \quad P(\theta_{\sigma(1)} = x_1, \ldots, \theta_{\sigma(m)} = x_m \mid s).$$

By de Finetti's theorem, exchangeability yields a mixture-of-i.i.d. representation (de Finetti 1929). We impose state symmetry and restrict the mixing variable to $q \in [1/2, 1]$, so every latent information environment is weakly informative. Let F denote the distribution of this common signal-accuracy variable. Conditional on s and q, signals are i.i.d. across voters and satisfy

$$P(\theta_i = s \mid s, q) = q$$

for every voter i and both states $s \in \{0,1\}$.

We assume that the de Finetti mixing distribution is non-vanishing in every right-neighbourhood of $\frac{1}{2}$. Thus, the common signal quality can be arbitrarily close to uninformative with positive probability. We call this the possibility of an uninformative environment. The signals are never systematically misleading: $q \geq 1/2$. Conditional on $q = 1/2$, different voters receive independent fair draws, not a single common random signal.

Assumption 1 (Possible uninformative environments)**.** *For every* $\eta > 0$*, the de Finetti mixing distribution satisfies*

$$F\left(\left[\frac{1}{2}, \frac{1}{2} + \eta\right]\right) > 0.$$

A pure strategy for each voter is a map $v_i : \{0,1\} \times \mathbb{R} \to \{0,1\}$, where $v_i(\theta_i, b_i)$ is the action of voter i after receiving signal θ_i and observing bias b_i. For the pure strategy calculations, select candidate 1 whenever a voter is indifferent. In the explicitly identified mixed equilibrium, allow randomisation between payoff equivalent actions, this relaxes the selection convention but not Bayesian optimality. Our equilibrium concept is that of Bayesian Nash equilibrium, which we refer to as equilibrium for short. Where explicitly considered below, mixed strategies allow independent private randomisation at each signal and bias.

3.1 Results

When a voter decides whether to vote for 0 or 1, they compare the payoff under these two actions given the actions of all other voters. A voter can influence their payoff

only when their vote can change the outcome of the election (i.e., when they are pivotal). This can happen if and only if candidates 0 and 1 are tied when counting the votes of the other N voters. Thus, a voter is pivotal if and only if exactly $\frac{N}{2}$ voters vote for 0 and $\frac{N}{2}$ voters vote for 1. Write $\Pi_i(v_{-i})$ for this event given the other voters' strategies, and suppress v_{-i} only when the profile is specified. For each realised signal $t \in \{0, 1\}$, define

$$\mu_i(t; v_{-i}) = P(s = 0 \mid \Pi_i(v_{-i}), \theta_i = t), \qquad r_i(t; v_{-i}) = P(s = t \mid \Pi_i(v_{-i}), \theta_i = t).$$

Thus $r_i(0) = \mu_i(0)$ and $r_i(1) = 1 - \mu_i(1)$, the two signal accuracy posteriors need not be equal for asymmetric strategy profiles. When pivotality has positive probability, the expected increase in utility from voting 0 rather than 1 is

$$\Delta u_i(t, b_i; v_{-i}) = 2P(\Pi_i(v_{-i}) \mid \theta_i = t)[\mu_i(t; v_{-i}) - \tfrac{1}{2} - b_i]. \tag{1}$$

Under this pure strategy selection, a voter votes 0 if and only if $\mu_i(t; v_{-i}) > 1/2 + b_i$, with strict inequality because indifferent voters choose 1. If pivotality has zero probability, both actions give the same payoff and the convention selects 1 without conditioning on a null event.

We say that a voter votes informatively, or follows their signal, if their vote coincides with their signal for both realisations. A voter votes following their bias if they vote 0 whenever $b_i < 0$ and 1 whenever $b_i > 0$, irrespective of the signal.

Our first result establishes that if voters vote informatively, then conditional on being pivotal, the probability that a voter's signal is correct tends to $\frac{1}{2}$ as the number of voters N increases.

Lemma 1. *Suppose Assumption 1 holds. If all voters other than i vote informatively, then for all $\varepsilon > 0$ there exists $\hat{N}(F, \varepsilon)$ such that if $N > \hat{N}$, then*

$$r_N := P(s = t \mid \Pi_i, \theta_i = t) \quad \in \quad \left[\frac{1}{2}, \frac{1}{2} + \varepsilon\right).$$

for both $t = 0$ and $t = 1$.

The proof of this lemma and all other proofs are in Appendix A. The intuition behind Lemma 1 is the following. Conditional on being pivotal and all other N voters voting informatively, the signals among the other voters must be split $\frac{N}{2}$ for candidate 0 and $\frac{N}{2}$ for candidate 1. By de Finetti's theorem, there exists a random variable q

such that, conditional on q, signals are i.i.d. and each is correct with probability q. Given that q is not known but its distribution is, the pivotal voter infers that, with a large number of voters, if they are pivotal, it must be that q is arbitrarily close to $\frac{1}{2}$, because an equal split has an exponentially smaller likelihood at any fixed $q > 1/2$ than sufficiently near $1/2$.

Given Lemma 1, we have the following best response result.

Proposition 1. *Suppose Assumption 1 holds. Fix any non-zero bias b_i. There exists $\hat{N}(F, |b_i|)$ such that for all $N > \hat{N}$, if all other voters vote informatively, then voting according to their bias is their unique best response.*

The intuition behind Proposition 1 is as follows. By Lemma 1, when all other voters vote informatively, the posterior probability that a voter's signal is correct approaches $1/2$ conditional on pivotality. Regardless of the sign or magnitude of any non-zero bias, the informational advantage of following the signal eventually becomes smaller than the payoff advantage of following the bias. Voting according to bias is therefore the unique best response.

The cutoff $\hat{N}$ is not uniform over biases arbitrarily close to zero. Nevertheless, signal following fails to be a Bayesian equilibrium for all sufficiently large N: choose any $\delta > 0$ for which $\gamma = G(\{|b| \geq \delta\}) > 0$. Once $r_N - 1/2 < \delta$, a set of types of probability at least γ has a strictly profitable deviation. If instead one fixes a realised, publicly known bias vector, the probability over independent draws that universal signal following survives is at most $(1-\gamma)^{N+1}$, which tends to zero. Small bias types can still follow their signals at any finite N, however.

In terms of the roles played by each assumptions, with independent signals of known accuracy $q_0 > 1/2$, the pivotal posterior remains q_0 when others follow signals. With random quality bounded below by an essential lower bound $q_{\min} \in (1/2, 1)$, the same argument gives $r_N \to q_{\min}$, so sufficiently small biases do not overturn signal following against that profile. Under Assumption 1, the limit is $1/2$. Without conditioning on pivotality, signal accuracy is $P(s = t \mid \theta_i = t) = E[q]$. In the symmetric informative profile the posterior distribution of q has weights proportional to $[4q(1-q)]^{N/2} dF(q)$. With zero bias and positive pivotal informativeness, following the signal still is a best response.

3.2 Equilibrium and electoral accuracy

We next construct a non-trivial equilibrium (such as those where all voters vote for one of the candidates) under an additional symmetry restriction on the bias distribution. This restriction is used only for this result, the finite examples later on have asymmetric bias distributions.

Proposition 2 (A responsive equilibrium and its limiting accuracy)**.** *Suppose Assumption 1 holds, $F((1/2,1]) > 0$, and the bias distribution G is atomless, symmetric about zero, and has full support on $[-B,B]$, where $B > 0$. Write*

$$\alpha(t) = G(t) - G(-t), \qquad d(q) = q - \tfrac{1}{2}.$$

For every even $N \geq 2$, there is a unique solution $t_N \in (0, E[q] - 1/2]$ to

$$t_N = \frac{\int d(q)\,[1 - 4\alpha(t_N)^2 d(q)^2]^{N/2}\,dF(q)}{\int [1 - 4\alpha(t_N)^2 d(q)^2]^{N/2}\,dF(q)}. \tag{2}$$

A Bayesian Nash equilibrium is to vote 0 after signal 0 if $b_i < t_N$, and to vote 0 after signal 1 if $b_i < -t_N$, voting 1 otherwise. Thus, apart from zero-probability types, voters with $|b_i| < t_N$ follow their signals and the others follow their biases.

Moreover, let $\alpha_N = \alpha(t_N)$ and let A_{N+1} be the ex ante probability that the elected candidate equals the state. Then

$$t_N \longrightarrow 0, \qquad \alpha_N \longrightarrow 0, \qquad \sqrt{N}\,\alpha_N \longrightarrow \infty, \qquad A_{N+1} \longrightarrow 1 - \tfrac{1}{2}F(\{1/2\}).$$

Conditional on (s,q), an equilibrium vote is correct with probability $1/2 + \alpha_N(q - 1/2)$. Although the proportion of signal following types vanishes, their aggregate effect dominates, and conditional on any $q > 1/2$ the correct candidate is selected with probability tending to one.

For the two-point information structure used below, write $d = \hat{q} - 1/2$. Equation (2) becomes the scalar equation

$$t_N = \frac{(1-p)d[1 - 4\alpha(t_N)^2 d^2]^{N/2}}{p + (1-p)[1 - 4\alpha(t_N)^2 d^2]^{N/2}}. \tag{3}$$

For example, if biases are uniform on $[-B, B]$, then $\alpha(t) = \min\{t/B, 1\}$, so the equilibrium is directly computable by one-dimensional root finding. Its exact finite-electorate accuracy is

$$A_{N+1} = \int \sum_{k=(N+2)/2}^{N+1} \binom{N+1}{k} [\tfrac{1}{2} + \alpha_N d(q)]^k [\tfrac{1}{2} - \alpha_N d(q)]^{N+1-k} \, dF(q). \tag{4}$$

3.3 Parameterised Example

In the experiment, we study the behaviour of a small ($N + 1 = 3$) as well as a mid-sized ($N + 1 = 15$) group of voters in the following parameterised setting. These examples replace the i.i.d. full-support bias draws by a fixed composition of one or five voters of each of three types, with the composition known. Candidate 1 corresponds to Orange and candidate 0 to Lemon. Appendix A.1 also constructs a mixed equilibrium requiring no designated roles within type. We fix a relatively simple correlated signal structure, where with probability $p = 0.1$ the environment is uninformative and each signal is an independent fair draw, and with probability $1 - p$ all signals have quality $\hat{q}$, that is,

$$q \;=\; \begin{cases} \frac{1}{2} \text{ with probability } p, \\ \hat{q} \text{ with probability } 1 - p, \end{cases}$$

where $\hat{q} \in \left(\frac{1}{2}, 1\right)$. Note the correlation structure: if the common signal quality is uninformative, then all voters' signals are uninformative. A useful quantity is the probability that a voter's own signal is correct given that the voter is pivotal and the other N voters are voting informatively:

$$r_N = P(s = t \mid \Pi_i, \theta_i = t) \;=\; \frac{p\frac{1}{2} + (1-p)\hat{q}(4\hat{q}(1-\hat{q}))^{\frac{N}{2}}}{p + (1-p)(4\hat{q}(1-\hat{q}))^{\frac{N}{2}}}. \tag{5}$$

Note that as N increases the quantity above converges to $\frac{1}{2}$. This is the result in Lemma 1. It means that, conditional on being pivotal, the posterior quality of the signals falls as the number of voters increases. Note also that r_N is non-monotonic in $\hat{q}$. That is, it is not true that increasing the ex ante signal quality increases the posterior belief about signal quality when pivotal. The reason is that when signal quality in the informative environment rises, pivotality becomes stronger evidence that the uninformative environment has occurred.

If $I \leq N+1$ is odd, a signal following voter faces $I-1$ other signal followers and an equal number of constant votes for the two candidates. Only in this balanced configuration is the common cutoff

$$b^*(I,\hat{q}) = r_{I-1} - \tfrac{1}{2} = \frac{(1-p)(\hat{q}-1/2)[4\hat{q}(1-\hat{q})]^{(I-1)/2}}{p+(1-p)[4\hat{q}(1-\hat{q})]^{(I-1)/2}}. \tag{6}$$

A voter strictly inside $(-b^*, b^*)$ follows their signal. At the boundaries, the exact interval for signal-following is $[-b^*, b^*)$ under our tie convention, provided $b^* > 0$. Voters outside this interval use a constant vote. To verify an equilibrium one must also check the incentives of the constant voters, who generally face a different pivotal event. Equations (A.1)-(A.2) give the posteriors for all these events, including asymmetric fixed votes.

Table 1 summarises thresholds b^* for the odd numbers of informative votes I in the balanced configuration and signal quality $\hat{q}$. For example, say we have three voters with biases $-0.2, 0.2$ and 0.3 respectively. If we have $\hat{q} = 0.9$, then for $I = 3$ we have $b^* = 0.305660$, which means that given that all three voters have their biases in $[-0.305660, 0.305660)$, there is an equilibrium where they all vote informatively. If, instead, we have $\hat{q} = 0.8$, then for $I = 3$ we have $b^* = 0.255621$, which, since the bias of voter 3 exceeds 0.255621, means that there is no equilibrium where all three voters vote informatively. For $I = 1$ we have $b^* = 0.27$, which means that if voters 1 and 3 vote according to their bias, then there is an equilibrium where voter 2 votes informatively since voter 2's bias is in $[-0.27, 0.27)$. If instead $\hat{q} = 0.95$, for $I = 3$ we have $b^* = 0.283948$. Therefore, since the bias of voter 3 is above 0.283948, there is again no equilibrium in which all three voters vote informatively. In this case, however, for $I = 1$ we have $b^* = 0.405$, which means that if voters 1 and 3 vote according to their bias, since voter 2's bias is in $[-0.405, 0.405)$, voter 2 votes informatively.

These three cases illustrate that the impact of signal quality on the maximum share of signal-following voters in pure equilibrium is not monotonic in this setting. In fact, increasing signal quality can reduce this maximum share, without necessarily reducing electoral accuracy. The reason is the non-monotonicity of the pivotal posterior described above.

Remark 1. *In this parameterised example, the maximum share of signal following voters in pure equilibrium is not monotonic in signal quality.*

	b^*		
I	$\hat{q} = 0.95$	$\hat{q} = 0.9$	$\hat{q} = 0.8$
15	0.000036	0.002801	0.085073
13	0.000190	0.007686	0.114640
11	0.001001	0.020644	0.147434
9	0.005217	0.052526	0.180476
7	0.026164	0.118291	0.210696
5	0.110352	0.215362	0.235985
3	0.283948	0.305660	0.255621
1	0.405000	0.360000	0.270000

Table 1: Balanced-pivot cutoff b^* for odd I and signal quality $\hat{q}$. Values are rounded to six decimal places, incentive checks use unrounded values.

We next consider the isolated impact of a pure increase in the number of voters. In particular, assume that there are 15 voters and the biases are as before, but each type (bias) exists five times. Therefore, the biases are

$$b = (-0.2, -0.2, -0.2, -0.2, -0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.3, 0.3, 0.3, 0.3, 0.3).$$

Consider first the case where $\hat{q} = 0.95$. If all 15 voters were informative ($I = 15$), we would have $b^* \simeq 0.000036$, which means that all voters would have incentives to vote according to their bias, a contradiction. The highest number of informative voters that can be sustained in a pure-strategy equilibrium is $I = 4$. One such equilibrium has all voters with bias -0.2 vote 0, all voters with bias 0.2 vote 1, four voters with bias 0.3 vote informatively, and the remaining voter with bias 0.3 vote 1. More generally, the four informative voters can be distributed across the two positive-bias types, with all remaining positive-bias voters voting 1. If instead $\hat{q} = 0.9$ or $\hat{q} = 0.8$, the highest number of informative voters that can be sustained in a pure-strategy equilibrium is $I = 5$: all voters with bias -0.2 vote 0, all voters with bias 0.3 vote 1, and all voters with bias 0.2 vote informatively. In these three parameterised cases, selecting a pure equilibrium with the maximum number of signal followers at each size, increasing the number of voters from 3 to 15 raises efficiency, defined as the probability that the majority vote correctly identifies the state, while weakly decreasing the share of voters who follow their signals. These calculations do not establish a general monotonicity result outside the parameterised examples.

Remark 2. *In these parameterised cases, selecting a pure equilibrium with the maximum number of signal followers, increasing the number of voters raises accuracy and*

weakly decreases the share of signal following voters. This is a selection-dependent comparison, not a statement about all equilibria.

Last, we consider the impact of a mean-preserving spread (MPS) in the bias distribution. Specifically, we now assume that the three bias levels are $(-0.5, 0.2, 0.6)$. For 3 voters ($N = 2$) with biases $-0.5, 0.2$ and 0.6, and irrespective of $\hat{q}$, an equilibrium is such that only the voter with bias 0.2 votes informatively. The voter with bias -0.5 votes for 0 for all signal values, and the voter with bias 0.6 votes for 1 for all signal values. For 15 voters such that each bias level is present five times, for $\hat{q} \in \{0.8, 0.9\}$ the maximum number of informative voters is 5, whereas if $\hat{q} = 0.95$ this number is 4. Hence, a mean-preserving spread in the bias distribution does not necessarily reduce the maximum number of signal-following voters in pure equilibrium. Both distributions have mean 0.1, their ordered partial sums are $-0.5 < -0.2$ and $-0.3 < 0$, with equal total 0.3, verifying the mean-preserving spread for equal type weights.

Remark 3. *In these finite examples, the stated mean-preserving spread does not necessarily decrease the maximum number of signal following voters in pure equilibrium.*

Finite predictions and efficiency benchmarks. Table 2 distinguishes best responses to universal signal following from equilibrium predictions. At 15 voters and $\hat{q} = .95$, for example, the former predicts bias following for every type, whereas the computed equilibria retain informative choices. The asymmetric four signal follower pure equilibrium has accuracy .943475, not .95, its state specific accuracies are .918633 in state 0 and .968317 in state 1. The symmetric within type mixed equilibrium in Appendix A.1 has accuracy .947654. None of these finite equilibria attains the full signal benchmark in this treatment. Among the eight experimental settings, only the baseline three-voter, $\hat{q} = .9$ case can attain the full-signal accuracy benchmark in equilibrium. As shown in Appendix A.1, attaining this benchmark requires universal signal following, which is not an equilibrium in the other seven settings. This statement also rules out exact attainment through mixing. The limiting 0.95 bound is attained by the equilibrium sequence in Proposition 2 under its additional assumptions, not established for the experimental composition.

Averaging the four rounds with equal weights gives full signal accuracies .912369 for 3 voters and .949031 for 15 voters. The corresponding averages for maximal signal following pure equilibria are .864950 and .931485 (or .932530 for 15 voters if the mixed

Round	n	$\hat{q}$	r_{n-1}	BR (A, B, C)	I^*	A_{pure}	A_{signals}
1	3	.90	.805660	(S, S, S)	3	.924800	.924800
1	15	.90	.502801	$(0, 1, 1)$	5	.942296	.949970
2	3	.80	.755621	$(S, S, 1)$	1	.770000	.856400
2	15	.80	.585073	$(0, 1, 1)$	5	.897872	.946184
3	3	.95	.783948	$(S, S, 1)$	1	.905000	.943475
3	15	.95	.500036	$(0, 1, 1)$	4	.943475	.950000
4	3	.90	.805660	$(0, S, 1)$	1	.860000	.924800
4	15	.90	.502801	$(0, 1, 1)$	5	.942296	.949970

Table 2: Exact finite-model predictions, rounded for display. Here $n = N + 1$, S means signal following, and $0, 1$ mean constant votes. BR gives a best response to all other voters following signals, not an equilibrium profile. The pivotal bias threshold is $r_{n-1} - 1/2$. I^* and A_{pure} refer to a pure equilibrium maximising the number of signal followers; A_{signals} is the accuracy under universal signal following. The displayed .950000 is .949999835 before rounding. Round 4 uses the spread distribution.

equilibrium is selected in round 3). These benchmarks measure state accuracy, not welfare including bias payoffs.

4 Experimental Design

In this section, we outline our experimental framework. The experiment is designed to assess the empirical relevance of the qualitative insights derived from the theory. The focus is less on comparing the experimental results with specific equilibrium predictions, although our design permits such comparisons, than on understanding qualitative patterns in participants' behaviour. Specifically, we are interested in understanding the extent to which participants understand pivotality and its implications for the quality of information.

4.1 Design

Our study consists of four distinct treatments in a 2×2 design, varying group size and whether or not the strategy method was employed. Participants were informed in the instructions whether they were in a small group ($N+1 = 3$) or in a large group

($N + 1 = 15$) and were explained the information structure, their decision tasks and how payoffs are calculated. Experimental instructions can be found in Appendix C. All treatments share the following common features.

Timing First, participants read the instructions and decided whether to accept or reject participation in this experiment. Following this, they engaged in a voting task. The task is to select one of two candidates, "Orange" or "Lemon", alongside other participants. At the beginning of each round, one of two possible states, orange or yellow, is randomly determined with equal probability. Candidate "Orange" is optimal in the orange state, whereas candidate "Lemon" is optimal in the yellow state. Participants receive information about the underlying state in the form of a signal—either an orange or a yellow ball. After observing the signal, they vote for either candidate.

In addition, participants have different preferences over the two candidates. There are three preference types (A,B and C), which are randomly assigned enforcing equal proportions. The personal parameters in favour of "Orange" or "Lemon" of each preference type are shown in Table 3. Participants' payoffs consist of three parts: (i) a fixed payment of GBP 1, (ii) GBP 1 if and only if the majority selects the correct candidate for the state, and (iii) the payoff associated with the personal parameter. For example, if the state is orange and a participant has a personal parameter of 0.2 in favour of "Lemon", a majority vote for "Orange" yields a payment of $1+1-0.2 = 1.8$ GBP. A majority vote for "Lemon" yields $1 + 0 + 0.2 = 1.2$ GBP.

A critical feature of the experiment is the reliability of these signals. With 90% probability, the round is in the informative environment, in which every participant's signal has accuracy $\hat{q} \in \{0.8, 0.9, 0.95\}$. Thus, even in the informative environment, a signal can be incorrect. With 10% probability, the round is in the uninformative environment, in which each signal is an independent fair draw and is therefore unrelated to the true state. Participants know the probability of each environment and the accuracy of informative signals before observing their signal in each round, but they do not observe which environment has been realised.

There are a total of four rounds in the experiment. Participants' personal parameters and the accuracy of informative signals vary from round to round. Table 3 depicts the details of our settings in each round. The four rounds appear in random order. At the beginning of each round, participants are informed of their preferences,

	Signal Accuracy $\hat{q}$	*Participants' Preferences*		
		Type A	Type B	Type C
round 1	0.90	-0.2	0.2	0.3
round 2	0.80	-0.2	0.2	0.3
round 3	0.95	-0.2	0.2	0.3
round 4	0.90	-0.5	0.2	0.6

Table 3: Signal accuracy and personal parameters. Signal accuracy gives the probability that the signal correctly reflects the true state in the informative environment. For participants' preferences, a negative sign denotes a preference for "Lemon", and the absolute value reflects the strength of the candidate preference.

the payments for each possible outcome, and the accuracy of informative signals. A picture of an orange or yellow ball is then presented as the signal, after which participants vote for a candidate. At the end of the experiment, two rounds were randomly selected for payment for each committee.[1] Because the matching in committees took place only at the end of the experiment, participants did not receive any feedback about other committee members' choices during the experiment.

Treatments Our treatments are designed to test the importance of two key ingredients underlying Proposition 1: conditional on being pivotal, the voter's signal (and those of other voters) becomes nearly uninformative as N grows. This approximation is sharp for sufficiently large N, but not when N is small. Our first treatment variation hence varies $N+1 \in \{3, 15\}$. Further, in order for this logic to work, voters need to be able to engage in contingent reasoning. They need to understand that the only situation that they ought to consider is the case where they are pivotal. In addition, they need to understand what pivotality implies for the likelihood that signals are informative. Our second treatment variation helps participants to condition on pivotality by explicitly asking them to do so in the strategy method. In these treatments (the SM treatments), we randomly match participants with voters from a previous session of the baseline treatments and ask them to make $N+1$ conditional decisions depending on how many of the voters they are matched with voted for "Orange" $(0, \ldots, N)$. In the rounds randomly selected for treatment they were then paid according to the relevant decision. Other than that, these treatments are identical to

[1] In the treatment with 15 participants we had some drop-outs which resulted in one committee having only 10 members. For this committee we randomly drew and then matched five participants from the remaining 270 participants twice, but we did not pay those participants for the second match.

the baseline treatments.

4.2 Procedures and Sample Characteristics

We conducted the online experiment in November 2024 with a sample of 1,123 UK residents registered on the survey platform Prolific. Median completion time ranged from 6:12 min in treatment 15 to 7:45 min in treatment SM15. Payments ranged from 0.50 GBP to 2.60 GBP with an average of 1.74 GBP. It was implemented in Qualtrics, and screenshots can be found in Appendix C. We received ethical approval from the Faculty Ethics Subcommittee 3 at the University of Essex under number ETH2324-2050. The experiment was pre-registered in the AEA RCT Registry under number AEARCTR-0014792. We pre-registered the basic design, outcomes and the planned sample size for the experiment. We had planned 270 participants per treatment and slightly over-recruited. Design and primary outcomes are unchanged compared to the pre-registration.

Table 4 reports sample characteristics. Sample characteristics were elicited at the end of the experiment. The questionnaire used to do so is shown in Appendix C. The table shows that the treatment samples are well balanced in terms of gender and age. Overall, participants are relatively highly educated and concentrated in the younger age groups.

5 Experimental Results

This section contains our experimental results. Section 5.1 summarizes some descriptive statistics for the different conditions before we discuss treatment comparisons in Section 5.2. Section 5.3 delves deeper into participants' understanding of pivotality and Section 5.4 contains an analysis using the post-experimental survey data.

5.1 Descriptive Statistics

Table 5 summarizes the pre-registered outcomes across our four treatment conditions. Efficiency measures the fraction of committee decisions that identify the correct state. For the SM treatments we base the analysis on synthetic committees by randomly

		3	15	SM3	SM15
Age	18-24	9.8%	10%	10.6%	9.6%
	25-34	29.8%	29.6%	25.7%	31.2%
	35-44	26.2 %	29.6%	31.7%	27.3%
	45-54	21.1%	14.6%	14.1%	18.7%
	55-64	8.0%	10%	13.7%	8.5%
	65+	5.1%	6.1%	4.2%	4.6%
Gender	Female	50.4%	49.3%	49.6%	51.1%
	Male	49.3%	49.6%	48.9%	48.9%
	Non-binary (third gender)	0.4%	0.4%	0.7%	0.0%
Education	Secondary school	14.9%	18.9%	14.5%	16.7%
	Vocational	16.7%	13.2%	17.3%	13.1%
	Some university	9.5%	7.5%	8.1%	8.9%
	Bachelor's degree	41.2%	37.5%	31.5%	37.2%
	Graduate degree	17.5%	21.4%	19.5%	21.6%
Risk tolerance		5.59	5.62	5.47	5.60
N		276	280	284	282
Committees		92	18	95	18

Table 4: Sample characteristics. Age: self-reported age bands. Gender: any shortfall from 100% is "Other". Education: highest level completed; any shortfall from 100% is "Prefer not to say" or did not complete secondary school. Risk tolerance: mean answer to question "On a scale from 0 to 10, where 0 means not at all willing to and 10 means extremely willing to, how willing are you to take risks in general?" One participant did not complete the demographic information.

matching 3 (or 15) participants from these treatments and considering their pivotal decisions. These measures hence tell us how efficiency would change if everyone had engaged in pivotal reasoning. Efficiency ranges from 77% to 83% across treatments and is somewhat higher in the larger elections. This seems reasonably high but is lower than what would be obtained under sincere voting on average ($\approx 91\%$ with 3 voters and $\approx 95\%$ with 15 voters). Correspondingly, welfare, measured by average payoffs, is highest in the elections with 15 voters. Even in the small committee, though, voters identify the correct state at least 77% of the time.

The fractions of participants who cast a vote that coincides with their signal also show some differences across treatments. Information aggregation is highest in the baseline treatment in the small committee and decreases both with the strategy method and as committee size increases. Across all treatments, more than 70% of

	Efficiency	*Signal Revelation*			*Welfare*
		Overall	(θ, b) align	(θ, b) misalign	
3	0.79	0.78	0.88	0.68	1.68 GBP
SM3	0.77	0.72	0.83	0.64	1.71 GBP
15	0.83	0.74	0.84	0.60	1.82 GBP
SM15	0.82	0.70	0.80	0.60	1.83 GBP

Table 5: Efficiency, information aggregation and welfare across treatments (pre-registered outcomes). Efficiency is measured by the fraction of committee decisions (majority votes) that identify the correct state. Signal revelation is the share of individual votes that coincide with the signal. Shares are reported overall and separately according to whether the signal θ is aligned with the direction of the bias, "(θ, b) align", or not. Welfare is the average payoff across all voters. In the SM treatments the "pivotal" vote is used for these computations.

participants vote in line with their signal and more than 80% do so when the signal points towards the state that they prefer in terms of their bias parameter b (the case "(θ, b) align").

5.2 Treatment Comparisons

We now study treatment comparisons more formally. Throughout this section, we rely on versions of the following linear probability model.

$$y_{ij}^t = \alpha + \beta_{15} T_{15} + \beta_{3SM} T_{\text{SM3}} + \beta_{15SM} T_{\text{SM15}} + \gamma X_i + \epsilon_{ij}^t. \tag{7}$$

y_{ij}^t is the outcome for individual i in group j in round t. In Table 7, it indicates whether the participant's vote matches their signal. T_{15}, T_{SM3}, and T_{SM15} are treatment indicators. X_i contains controls for individual characteristics.

To compare efficiency across treatments, i.e., whether the majority vote identified the state correctly (efficiency, Table 6), we collapse the data at the group level and run a version of equation (7) without the individual-level controls X_i.

Table 6 shows regression results for our efficiency measure. We first focus on the impact of increasing the number of voters, measured by β_{15}. Column (1) uses all data aggregated across different realisations of the bias distribution and $\hat{q}$. Efficiency is higher in the treatments with more voters than in the baseline, consistent with the parameterised examples in Section 3.3, although the difference is not statistically

significant. Columns (2)–(4) then split the data by $\hat{q}$, keeping the bias distribution constant. Efficiency is higher in the larger groups for $\hat{q} \leq 0.9$ and lower for $\hat{q} = 0.95$. The directions for $\hat{q} \leq 0.9$ are in line with the parameterised examples, whereas the estimate for $\hat{q} = 0.95$ moves in the opposite direction. Only the estimate for $\hat{q} = 0.8$ is marginally significant at the 10% level. Finally, column (5) singles out the case with the high-variance bias distribution; here, too, efficiency is higher in the larger group, although the difference is not statistically significant. We next consider the impact of the strategy method. Across group sizes and signal accuracies, the estimates do not provide statistically significant evidence that the strategy method changes efficiency (β_{3SM} and comparison of β_{15} and β_{15SM}, respectively).

	Efficiency				
	(1)	(2)	(3)	(4)	(5)
	All	$\hat{q} = 0.9$	$\hat{q} = 0.95$	$\hat{q} = 0.8$	MPS
15 treatment (β_{15})	0.034	0.046	-0.191	0.174*	0.139
	(0.053)	(0.093)	(0.117)	(0.095)	(0.107)
SM3 treatment (β_{3SM})	-0.022	-0.028	-0.015	-0.060	0.016
	(0.030)	(0.054)	(0.067)	(0.055)	(0.060)
SM15 treatment (β_{15SM})	0.043	-0.005	-0.033	0.068	0.145
	(0.051)	(0.093)	(0.117)	(0.095)	(0.104)
Constant	0.785***	0.848***	0.717***	0.826***	0.750***
	(0.021)	(0.038)	(0.048)	(0.039)	(0.043)
Observations	896	224	224	224	224
R-squared	0.003	0.003	0.012	0.030	0.016

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 6: Efficiency regressed on treatment dummies. Simple OLS regression. Baseline category is the $N + 1 = 3$ treatment.

Table 7 shows treatment comparisons for the share of participants who truthfully reveal their signal. Columns (1) and (2) show all data, where column (2) includes controls for individual characteristics. The table shows that information aggregation is lower in the larger committee: a smaller fraction of participants vote in line with their signal. This is true on average and (in line with the theoretical prediction) is largely driven by the case with the highest signal accuracy $\hat{q} = 0.95$ and the MPS case (columns (4) and (6)). Averaged across conditions, efficiency is nevertheless slightly higher in the larger committee (Table 6, column (1)), although the accuracy-specific estimates are heterogeneous. The parameterised examples illustrate how the number of informative votes can increase even when their share weakly decreases. The table

	Signal Revelation					
	(1)	(2)	(3)	(4)	(5)	(6)
	All	All	$\hat{q} = 0.9$	$\hat{q} = 0.95$	$\hat{q} = 0.8$	MPS
15 treatment (β_{15})	-0.045**	-0.042**	-0.014	-0.075**	-0.008	-0.073**
	(0.019)	(0.019)	(0.035)	(0.036)	(0.035)	(0.036)
SM3 treatment (β_{3SM})	-0.059***	-0.058***	-0.051	-0.084**	-0.041	-0.055
	(0.020)	(0.020)	(0.036)	(0.036)	(0.036)	(0.036)
SM15 treatment (β_{15SM})	-0.081***	-0.083***	-0.053	-0.129***	-0.038	-0.111***
	(0.019)	(0.019)	(0.036)	(0.037)	(0.036)	(0.037)
Constant	0.788***	0.847***	0.858***	0.791***	0.841***	0.899***
	(0.012)	(0.039)	(0.068)	(0.071)	(0.068)	(0.068)
Observations	4,492	4,440	1,110	1,110	1,110	1,110
Demographics	No	Yes	Yes	Yes	Yes	Yes
R-squared	0.2075	0.2244	0.094	0.102	0.050	0.135
Individuals	1,123	1,110	1,110	1,110	1,110	1,110

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 7: The share of participants voting with their signal regressed on treatment dummies. Random effects panel regression (columns (1)-(2)), simple OLS (columns (3)-(6)). Demographics include a gender dummy, age, education categories and risk attitude. Baseline category is the $N + 1 = 3$ treatment. Standard errors clustered at the individual level.

also shows that, on average, fewer votes coincide with signals under the strategy method. One possible interpretation is that encouraging participants to think about pivotality reduces unstrategic reliance on their signals.

Our last table of treatment comparisons focuses on the impact of signal accuracy (Table B.1). The estimates provide some evidence that lower signal accuracy reduces information aggregation. The higher signal accuracy ($\hat{q} = 0.95$), however, is not statistically different from the baseline $\hat{q} = 0.9$.

5.3 The role of pivotality

In this section, we focus in more detail on the SM treatments. These treatments fulfil two roles. First, they allow us to study the extent to which failure to condition on pivotality is responsible for our results. Second, they allow us to observe the full strategies (response functions) of our participants.

Recall that the main theoretical mechanism behind Proposition 1 requires agents first to reason about pivotality, as they need to understand that their vote matters

if and only if they are pivotal. Second, it requires them to have an intuitive understanding of the law of large numbers, as they need to realise that conditional on being pivotal it becomes increasingly less likely that signals are informative as the number of voters increases.

Pairwise tests indicate that participants are more likely to vote with their bias when they know they are pivotal, for both $N+1=3$ and $N+1=15$ (3 vs. SM3, two-sided t-test $p<0.01$; 15 vs. SM15, $p<0.01$). For the smaller group, the equilibrium comparison depends on the round: the most informative equilibrium has all three voters follow their signals only in the $\hat{q}=0.9$ round with the baseline bias distribution; in the other parameterised rounds it has one informative voter. The pooled SM3 effect therefore has no single equilibrium-direction interpretation. For the larger group ($N+1=15$), the most informative pure-strategy equilibria in our parameterised example involve 27–33% of participants voting with their signal, depending on signal accuracy (see Section 3.3); the strategy method therefore moves behaviour in the direction of those equilibria.

Despite this, the impact of group size is not more pronounced when people know they are pivotal. For the 3- and 15-voter baseline treatments, the decline in the fraction of people voting with their signal is about 0.045 (coefficient β_{15} in Table 7). In the SM treatments, by contrast, the corresponding decline is only $\beta_{15SM}-\beta_{3SM}=-0.081-(-0.059)\approx-0.022$.

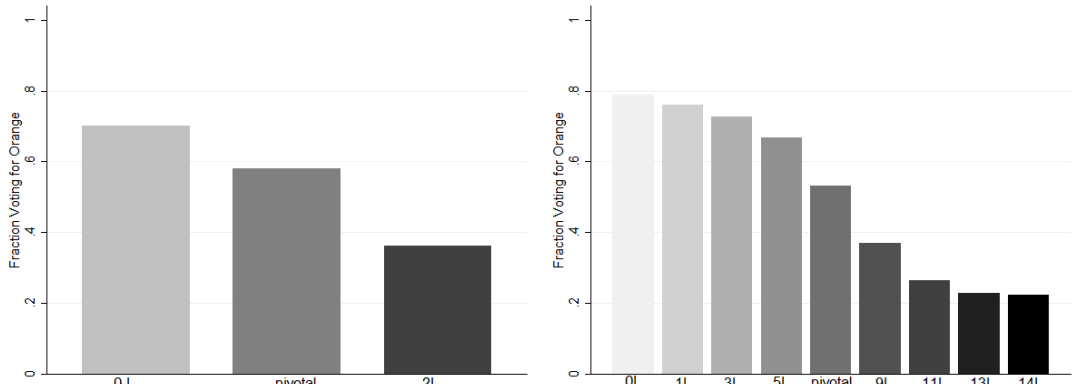


Figure 1: Response functions in SM3 (left panel) and SM15 (right panel). The figure shows the fraction of participants voting for candidate "Orange" depending on the number of voters voting for "Lemon".

Beyond participants' pivotal decisions, the SM treatments reveal their full response functions. Figure 1 shows that on average participants become less likely to vote for

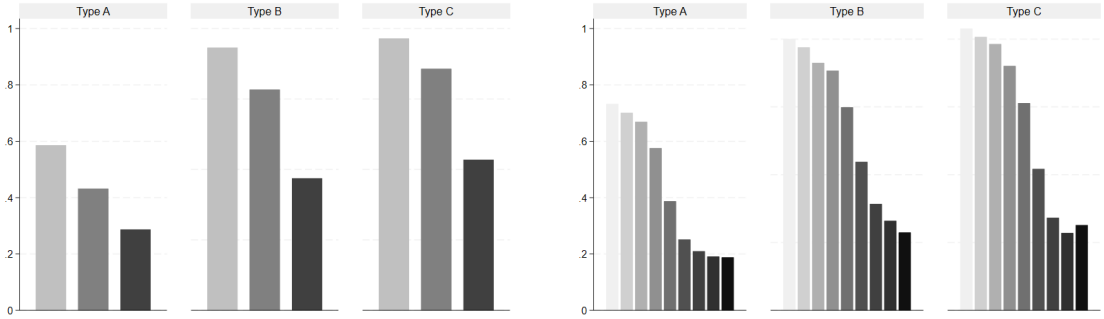


Figure 2: Response functions by individual bias in SM3 (left panel) and SM15 (right panel). The figure shows the fraction of participants voting for candidate "Orange" depending on the number of voters voting for "Lemon".

"Orange" as more of the other voters choose "Lemon". The response is monotone and approximately smooth through the pivotal event, rather than displaying the asymmetric break implied by a sharp retreat to bias at pivotality. Choices away from the pivotal event have no effect on the election outcome, so these responses should be interpreted as revealing participants' contingent decision rules rather than payoff-relevant best responses. Figure 2 reports response functions for the three bias types. As expected, participants with a stronger bias towards "Orange" are more likely to vote for "Orange". Beyond this level effect, the shape of the aggregate response function is similar across types.

Strategy	**SM3**	**SM15**
Constant	61%	45%
Always Bias	28%	14%
Always Signal	35%	19%
Responsive	39%	55%
Follow Majority	39%	0%

Table 8: Distribution of individual response functions in treatments SM3 and SM15 at the individual-round level. Constant indicates the fraction of response functions who always vote "Orange" or always vote "Yellow". Always Bias is a constant response following the bias and Always Signal is a constant response function following the signal. Responsive response functions are not constant. Follow Majority are responses that always vote in line with the majority and allow for any vote in the pivotal case.

We can also study individual response functions. Table 8 shows the distribution of individual response functions. In treatment SM3 around 61% of individual response functions are constant, while 39% are responsive to the fraction of other voters who

vote orange. All of the latter follow the majority, i.e. vote orange when both others vote orange and yellow when both others vote yellow. Out of the constant response functions similar proportions always follow the bias or always follows the signal. (Note that these need not add up to 61% as bias and signal can coincide.) In treatment SM15 where participants have to indicate more contingent choices, fewer responses are constant ($\approx$ 45%) and correspondingly more are responsive ($\approx$ 55%). None of the latter always follow the majority, but most of them follow the majority in most of the indicated contingent choices.

On balance analysis of participants' response functions shows little evidence that people reason about the informativeness implications of being pivotal. This is the case despite the fact that we explicitly ask about the pivotal decision. Note also that even if participants did understand the theoretical informativeness implications of a tied vote this connection is substantially weakened in the experiment as a substantial fraction of participants do not vote with their signal.

5.4 Self-reported Reasoning

To gain a better understanding of participants' reasoning, we asked them at the end of the experiment which factors they were thinking about when making their decisions. We gave them four answer options and also allowed them to describe additional considerations in a free-form field.

Figure 3 shows the results. In our baseline treatments about 10–15 percent of participants indicate that pivotality was the key factor they were thinking about. This share increases to $\approx$ 20 percent in the SM treatments (two-sided t-test, $p < 0.05$). Hence, asking them to make a conditional choice does indeed seem to make the question of pivotality more salient to them and induces participants to think more about it.

Around 10–20 percent of participants report mainly thinking about how informative their signal is. Interestingly, this percentage is somewhat higher in the smaller group. In the large committee, only the SM treatment makes them think more about others and less about the tension between their signal and preference bias ("bias-info tradeoff"). There are no additional factors reported by more than 5 percent of participants via the free-form field.

These results show that participants pay attention, or at least claim to pay at-

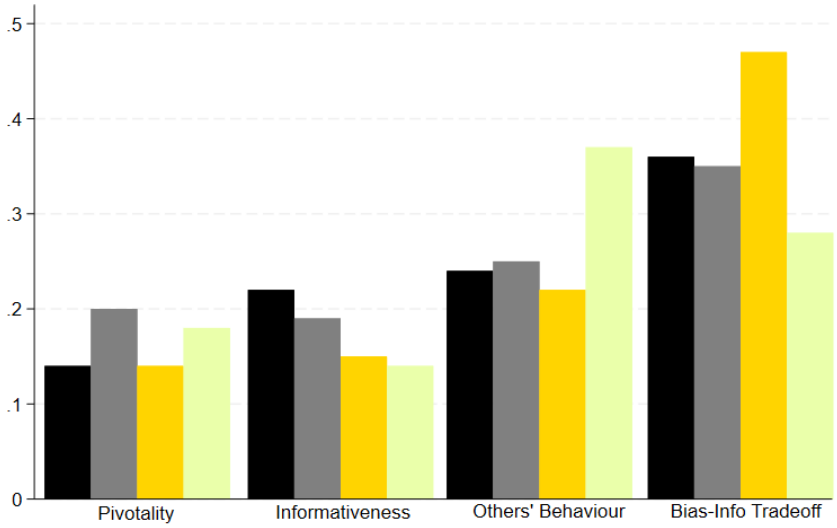


Figure 3: Fraction of participants who report thinking mainly about (i) pivotality, (ii) informativeness of the signals, (iii) others' behaviour or (iv) their own bias-information tradeoff in the experiment across treatments. Black bars to the left in each category are treatment $N+1=3$, grey bars (centre-left) treatment SM3, dark yellow bars (centre-right) treatment $N+1=15$ and light yellow bars (right in each category) treatment SM15. The remaining responses are classified as "Other". The exact statements can be found in Appendix C.

tention, to theoretically relevant considerations when making their choice. However, they also show that less than a fifth of participants report thinking about pivotality even when explicitly prompted.

6 Discussion and Conclusion

Predictions of voting models often hinge on contingent reasoning: the fact that people realise that the consequences of their actions depend on a contingent event and therefore reason as if the event actually took place. In voting theory, the contingent event is being pivotal. In common-interest environments, this type of reasoning is often used to connect naive information-aggregation arguments to equilibrium ones; for example, McLennan (1998) shows that beneficial information aggregation can be supported in equilibrium even when equilibrium strategies need not coincide with sincere or informative voting. Our setting departs from this benchmark in two ways: preferences are heterogeneous and private, and information is correlated, with a small probability of a collapse in informativeness. The mechanism behind Proposition 1 re-

quires not only conditioning on pivotality, but also using pivotality to infer that the realised informativeness of signals is likely to be low (Lemma 1), a logic that is absent from standard independent-signal models and that is sharpened by exchangeability and uninformative signals.

Our experiment suggests that this second step is behaviourally weak. In the strategy-method treatments, pivotality is made explicit, so failure to reduce the problem to the pivotal case is less of a concern. Nevertheless, response functions do not display the sharp retreat to bias predicted by the pivot-based mechanism. This pattern fits a broader body of evidence that people often struggle with state-by-state and hypothetical conditioning in strategic environments (Esponda & Vespa 2014, Martínez-Marquina et al. 2019, Moser 2019, Niederle & Vespa 2023) and with related behavioural models such as cursed equilibrium (Eyster & Rabin 2005). In our context, a parsimonious interpretation is that many participants underweight correlation when they interpret the vote split, so a close election is not treated as strong evidence of a breakdown in informativeness. This aligns with the correlation-neglect literature documenting insufficient discounting of redundant information from common sources (DeMarzo et al. 2003, Glaeser & Sunstein 2009, Ortoleva & Snowberg 2015, Enke & Zimmermann 2019, Hossain & Okui 2024) and with evidence that correlation can lead to overreaction when it is not properly internalised (Moser & Wallmeier 2021). While Moser & Wallmeier (2021) vary correlation across treatments, our treatment design was focused on isolating the impact of pivotal reasoning. Our results show that failure to reduce to the pivotal case is not a main driver of the result. Overall, the experiment does not support the comparative statics generated by fully rational pivotal inference under correlated signals. The theoretical benchmark nevertheless clarifies when correlation can be decisive, while the data indicate that its force depends on whether voters recognise and use the correlation structure.

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A Proofs

Proof of Lemma 1. By de Finetti's theorem, state symmetry and Bayes' rule, when all other voters vote informatively, for either realised signal t,

$$\begin{aligned} r_N &= \frac{\int_{1/2}^1 q^{\frac{N}{2}+1}(1-q)^{\frac{N}{2}}\,dF(q)}{\int_{1/2}^1 q^{\frac{N}{2}}(1-q)^{\frac{N}{2}}\,dF(q)} \\ &= \frac{\int_{1/2}^1 q\,[4q(1-q)]^{\frac{N}{2}}\,dF(q)}{\int_{1/2}^1 [4q(1-q)]^{\frac{N}{2}}\,dF(q)}. \end{aligned}$$

Because $q \in [1/2, 1]$, the ratio is at least $1/2$. It remains to show that it is eventually below $1/2 + \varepsilon$.

Fix any $\rho \in (0, 1/2)$. Let $B_\rho = [1/2, 1/2 + \rho]$ and $A_\rho = (1/2 + \rho, 1]$. Then

$$r_N \leq \frac{1}{2} + \rho + \frac{\int_{A_\rho}[4q(1-q)]^{\frac{N}{2}}\,dF(q)}{\int_{B_\rho}[4q(1-q)]^{\frac{N}{2}}\,dF(q)}.$$

On A_ρ, $4q(1-q) \leq 1 - 4\rho^2$. Therefore

$$\int_{A_\rho}[4q(1-q)]^{\frac{N}{2}}\,dF(q) \leq (1-4\rho^2)^{\frac{N}{2}}.$$

By Assumption 1, $F([1/2, 1/2+\rho^2]) > 0$. On $[1/2, 1/2+\rho^2]$, $4q(1-q) \geq 1 - 4\rho^4$, so

$$\int_{B_\rho}[4q(1-q)]^{\frac{N}{2}}\,dF(q) \geq F([1/2, 1/2+\rho^2])(1-4\rho^4)^{\frac{N}{2}}.$$

Thus

$$\frac{\int_{A_\rho}[4q(1-q)]^{\frac{N}{2}}\,dF(q)}{\int_{B_\rho}[4q(1-q)]^{\frac{N}{2}}\,dF(q)} \leq \frac{1}{F([1/2, 1/2+\rho^2])}\left(\frac{1-4\rho^2}{1-4\rho^4}\right)^{\frac{N}{2}}.$$

The term in parentheses is strictly smaller than one. Hence the ratio converges to zero as $N \to \infty$. Choose $\rho < \varepsilon/2$ and then choose N sufficiently large that the last ratio is below $\varepsilon/2$. The result follows. □

Proof of Proposition 1. Fix a voter with $b_i \neq 0$ and choose $\varepsilon \in (0, |b_i|)$. By Lemma 1, there exists $\hat{N}(F, |b_i|)$ such that, for all $N > \hat{N}$,

$$r_N < \frac{1}{2} + \varepsilon.$$

If $b_i < 0$, then after signal $\theta_i = 0$ the condition for voting 0 is

$$r_N - \frac{1}{2} - b_i > 0,$$

which holds strictly because $r_N \geq 1/2$ and $b_i < 0$. After signal $\theta_i = 1$, the condition for voting 0 is

$$1 - r_N - \frac{1}{2} - b_i > 0.$$

The left-hand side is greater than $-\varepsilon - b_i > 0$. Hence a voter with $b_i < 0$ strictly prefers voting 0 after both signals.

If $b_i > 0$, then after signal $\theta_i = 0$ the condition for voting 0 satisfies

$$r_N - \frac{1}{2} - b_i < \varepsilon - b_i < 0,$$

so the voter strictly prefers voting 1. After signal $\theta_i = 1$, the condition for voting 0 is also violated because

$$1 - r_N - \frac{1}{2} - b_i \leq -b_i < 0.$$

Hence a voter with $b_i > 0$ strictly prefers voting 1 after both signals. Therefore voting according to bias is the unique best response for all $N > \hat{N}$. □

Proof of Proposition 2. For a given t, symmetry and independence of biases imply that, conditional on (s, q), each other voter selects the correct candidate with probability $r(q, t) = 1/2 + \alpha(t)(q - 1/2)$. The pivotal likelihood is the same in both states and is proportional to $[1 - 4\alpha(t)^2(q - 1/2)^2]^{N/2}$. Bayes' rule therefore gives the posterior signal-accuracy advantage on the right-hand side of (2). Calling this quantity $H_N(t)$, the best-response thresholds are $H_N(t)$ after signal 0 and $-H_N(t)$ after signal 1.

The function H_N is continuous, its denominator is positive because F has mass arbitrarily near $1/2$. Moreover, $H_N(0) = E[q] - 1/2 > 0$, and $H_N(t) \leq E[q] - 1/2$. To see monotonicity, if $0 \leq \alpha_1 < \alpha_2 \leq 1$, the ratio

$$\left(\frac{1 - 4\alpha_2^2 d^2}{1 - 4\alpha_1^2 d^2}\right)^{N/2}$$

is non-increasing in $d \in [0, 1/2]$ wherever the denominator is positive. Increasing α thus shifts the posterior weight towards lower q, so $H_N(t)$ is non-increasing in t.

Continuity and the strictly increasing left-hand side of $t = H_N(t)$ give existence and uniqueness of t_N. The stated strategy follows from the best-response rule, including the specified treatment of indifferent types.

For every fixed $\epsilon > 0$, full support gives $\alpha(\epsilon) > 0$. The concentration argument in Lemma 1, with weights $[1 - 4\alpha(\epsilon)^2 d^2]^{N/2}$, gives $H_N(\epsilon) \to 0$. If $t_N \geq \epsilon$ along a subsequence, then $t_N = H_N(t_N) \leq H_N(\epsilon) \to 0$, a contradiction. Hence $t_N \to 0$ and, by atomlessness, $\alpha_N \to 0$.

Suppose along a subsequence that $N\alpha_N^2 \leq K < \infty$. Since $4d^2 \leq 1$, for all sufficiently large N on that subsequence,

$$[1 - 4\alpha_N^2 d^2]^{N/2} \geq (1 - \alpha_N^2)^{N/2} \geq e^{-K}.$$

(The last inequality follows from $\log(1-x) \geq -2x$ for small x.) The numerator in (2) is then at least $e^{-K} E[q - 1/2] > 0$ and its denominator is at most one, contradicting $t_N \to 0$. Thus $N\alpha_N^2 \to \infty$.

Conditional on any $q > 1/2$, the $N + 1$ correctness indicators are independent Bernoulli variables with mean $1/2 + \alpha_N(q - 1/2)$. Chebyshev's inequality bounds the probability of an incorrect majority by $[4(N+1)\alpha_N^2(q-1/2)^2]^{-1}$, which tends to zero. At $q = 1/2$, accuracy is exactly $1/2$. Dominated convergence proves the limit in the proposition. Conditional on $q = 1/2$, states are equiprobable and the entire vector of signals and biases is independent of the state. Any rule based on these observations therefore has accuracy $1/2$ there and at most one elsewhere. This proves the upper bound and its asymptotic attainment. □

A.1 Finite equilibrium calculations

Posteriors for arbitrary strategies. For any opponents' strategy profile, let $L_{i,s}(q)$ be the probability of exactly $N/2$ votes for candidate 1 among the other N voters, conditional on (s, q). Let $\ell_s(t, q) = q$ if $t = s$ and $\ell_s(t, q) = 1 - q$ otherwise. Provided the denominator is positive,

$$\mu_i(t; v_{-i}) = \frac{\int L_{i,0}(q)\ell_0(t, q)\, dF(q)}{\int [L_{i,0}(q)\ell_0(t, q) + L_{i,1}(q)\ell_1(t, q)]\, dF(q)}. \tag{A.1}$$

This formula also allows mixed strategies and uncertainty about others' strategies. Under the independent bias draws of the main model, set

$$a_j(s, q) = \int [qv_j(s, b) + (1 - q)v_j(1 - s, b)]\, dG(b),$$

where $v_j(t, b)$ now denotes the probability of voting 1. Then $L_{i,s}(q)$ is the coefficient of $z^{N/2}$ in $\prod_{j \neq i}[1 - a_j(s, q) + a_j(s, q)z]$.

Pure strategies and asymmetric fixed votes. Suppose among i's opponents there are m signal followers, h voters always voting 1, and $N - m - h$ always voting 0. Pivotality requires $k = N/2 - h$ signals equal to 1 among the m signal followers. For $0 \leq k \leq m$,

$$L_{i,0}(q) = \binom{m}{k}(1-q)^k q^{m-k}, \qquad L_{i,1}(q) = \binom{m}{k} q^k (1-q)^{m-k}. \tag{A.2}$$

If k is outside this range, pivotality is impossible and the indifference convention selects 1 after both signals. Otherwise define $D_i(t) = \mu_i(t; v_{-i}) - 1/2 - b_i$. The complete incentive checks are

strategy	$t = 0$	$t = 1$
0	$D_i(0) > 0$	$D_i(1) > 0$
S	$D_i(0) > 0$	$D_i(1) \leq 0$
1	$D_i(0) \leq 0$	$D_i(1) \leq 0$

Here S denotes signal following. At a positive probability pivotal event, $\mu_i(0) \geq \mu_i(1)$: writing the four joint likelihoods as $A = \int q L_{i,0}\, dF$, $B = \int (1-q) L_{i,0}\, dF$, $C = \int q L_{i,1}\, dF$ and $D = \int (1-q) L_{i,1}\, dF$, we have $A \geq B$, $C \geq D$, and hence $AC \geq BD$. Therefore $A/(A+D) \geq B/(B+C)$. Voting against both signals cannot be a best response under our tie convention.

For $n = N + 1$ voters, with I signal followers and h constant votes for 1, define $T(I, r, x) = P(\text{Bin}(I, x) \geq r)$, with $T = 1$ for $r \leq 0$ and $T = 0$ for $r > I$. Exact electoral accuracy is

$$A(n, I, h) = \frac{1}{2} \int \left[1 - T\left(I, \frac{n+1}{2} - h, 1 - q\right) + T\left(I, \frac{n+1}{2} - h, q\right) \right] dF(q). \tag{A.3}$$

For the accuracy, the full signal benchmark is $A(n, n, 0)$. It is the optimal decision rule based on all n signals: a majority for 1 has likelihood at least as high in state 1 as in state 0 at every $q \geq 1/2$. Because $\hat{q} \in (1/2, 1)$ and $p < 1$, every signal vector has positive probability and the full-signal majority decision is uniquely optimal. To implement it, fix any voter and hold the other signals at an equal split. The correct full signal decision switches from 0 to 1 with that voter's signal, so that voter must cast

0 after signal 0 and 1 after signal 1 with probability one. Applying this to each voter shows that only universal signal following, including among mixed profiles, attains the benchmark. This statistical criterion is distinct from maximising average utility. For a fixed bias vector with mean $\bar{b}$, expected average utility is $A + (2P(c = 1) - 1)\bar{b}$, add the experimental fixed payment of one if reporting monetary welfare.

Enumeration and equilibrium selection. For each type, enumerate all triples (n_0, n_S, n_1) summing to the number of that type. With one of each type there are $3^3 = 27$ count configurations; with five of each type there are $21^3 = 9{,}261$. For each occupied type–strategy cell, remove that voter, compute (A.2) and (A.1), and check both signal realisations. The accompanying script uses exact rational arithmetic for these checks. Table A.1 gives all surviving configurations up to permutations within type, including the unresponsive equilibrium. Thus the maximality claims concern all pure equilibria respecting the stated indifference convention, not merely the equilibria with constant votes split evenly. They do not bound mixed-equilibrium accuracy.

Biases	n	$\hat{q}$	All responsive pure equilibria
Baseline	3	.8	$(0, S, 1)$
Baseline	3	.9	(S, S, S), $(0, S, 1)$, $(0, 1, S)$
Baseline	3	.95	$(0, S, 1)$, $(0, 1, S)$
Baseline	15	.8, .9	$(0^5, S^5, 1^5)$
Baseline	15	.95	$(0^5, S^j 1^{5-j}, S^{4-j} 1^{1+j})$, $j = 0, \ldots, 4$
Spread	3	.8, .9, .95	$(0, S, 1)$
Spread	15	.8, .9	$(0^5, S^5, 1^5)$
Spread	15	.95	$(0^5, S^4 1, 1^5)$

Table A.1: All responsive pure equilibria in the finite examples. Entries give the strategies of types (A, B, C). Superscripts denote numbers of voters, not powers. The all-1 equilibrium is present in every case in addition to the listed profiles. Baseline biases are $(-.2, .2, .3)$; spread biases are $(-.5, .2, .6)$. Role assignments are known in the asymmetric profiles.

$\hat{q}$	Type	Strategy	(m, k)	$\mu_i(0)$	$\mu_i(1)$	$1/2 + b_i$
.8	A	0	$(5, 2)$	.867812	.500000	.300000
.8	B	S	$(4, 2)$	.735985	.264015	.700000
.8	C	1	$(5, 3)$	.500000	.132188	.800000
.9	A	0	$(5, 2)$	.820341	.500000	.300000
.9	B	S	$(4, 2)$	.715362	.284638	.700000
.9	C	1	$(5, 3)$	.500000	.179659	.800000
.95	A	0	$(4, 1)$	.927045	.610352	.300000
.95	B	1	$(4, 2)$	.610352	.389648	.700000
.95	C	S	$(3, 1)$	.875815	.500000	.800000
.95	C	1	$(4, 2)$	.610352	.389648	.800000

Table A.2: Both-signal incentive checks for representative maximal-signal-following equilibria with 15 voters and baseline biases. Vote 0 precisely when the relevant posterior exceeds the last column. For .95, the representative has four informative C-types. The remaining configurations, three voter cases and spread cases are verified by the same formula and reported in the accompanying calculation files.

An equilibrium without designated roles within type. In the 15-voter, $\hat{q} = .95$ example, the pure profiles with four signal followers distinguish voters of the same type. Allowing mixing at payoff indifference, an alternative Bayesian equilibrium uses the same strategy for every member of a type and requires only the fixed type counts. Let A-types always vote 0, C-types always vote 1, and each B-type vote 0 with probability x after signal 0, and vote 1 after signal 1, using independent private randomisation. Since the A and C votes cancel, a B-voter is pivotal exactly when the other four B-votes are split equally. For $x > 0$, indifference after signal 0 is

$$\frac{\int q^3(1-xq)^2\,dF(q)}{\int \{q^3(1-xq)^2 + (1-q)^3[1-x(1-q)]^2\}\,dF(q)} = .7. \tag{A.4}$$

For $p = .1$ and $\hat{q} = .95$, a solution is $x = .9709159562$. The pivotal state posteriors are

type	$\mu_i(0)$	$\mu_i(1)$
A	.785339	.518655
B	.700000	.403947
C	.534178	.326988

Thus every type's incentives are satisfied for both baseline and spread biases. Accuracy is .947654, with state specific accuracies .943540 and .951769 for states 0 and 1, respectively. This exceeds the best pure-equilibrium accuracy .943475 but remains below the full signal benchmark .949999835. This construction is not an enumeration of all mixed equilibria.

B Additional Tables

	Information Aggregation		
	(1)	(2)	(3)
$\hat{q} = 0.8$	-0.034**	-0.034*	-0.009
	(0.017)	(0.017)	(0.024)
$\hat{q} = 0.95$	0.008	0.009	0.005
	(0.017)	(0.017)	(0.024)
Large Committee			0.000
			(0.025)
$\hat{q} = 0.8\times$ Large Committee			-0.008
			(0.035)
$\hat{q} = 0.95\times$ Large Committee			-0.058*
			(0.034)
Constant	0.748***	0.795***	0.801***
	(0.012)	(0.042)	(0.045)
Observations	3,369	3,330	3,330
Demographics	No	Yes	Yes
R-squared	0.002	0.004	0.005
Individuals	1,123	1,110	1,110

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table B.1: Information aggregation regressed on signal accuracy (baseline is $\hat{q} = 0.9$), a dummy for the large committee ($N + 1 = 15$), and interactions. Data only from rounds with the baseline bias distribution. Demographics include a gender dummy, age, education categories, and risk attitude.

C Experimental Screenshots

Figure C.1: Welcome and Consent

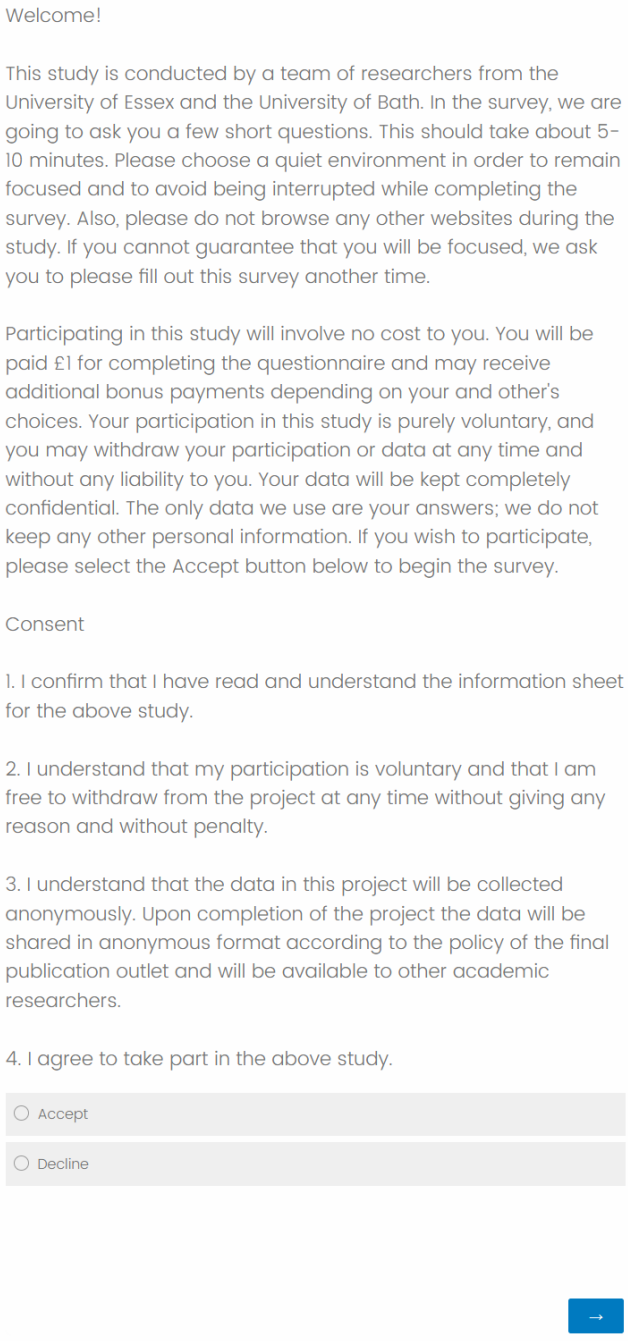
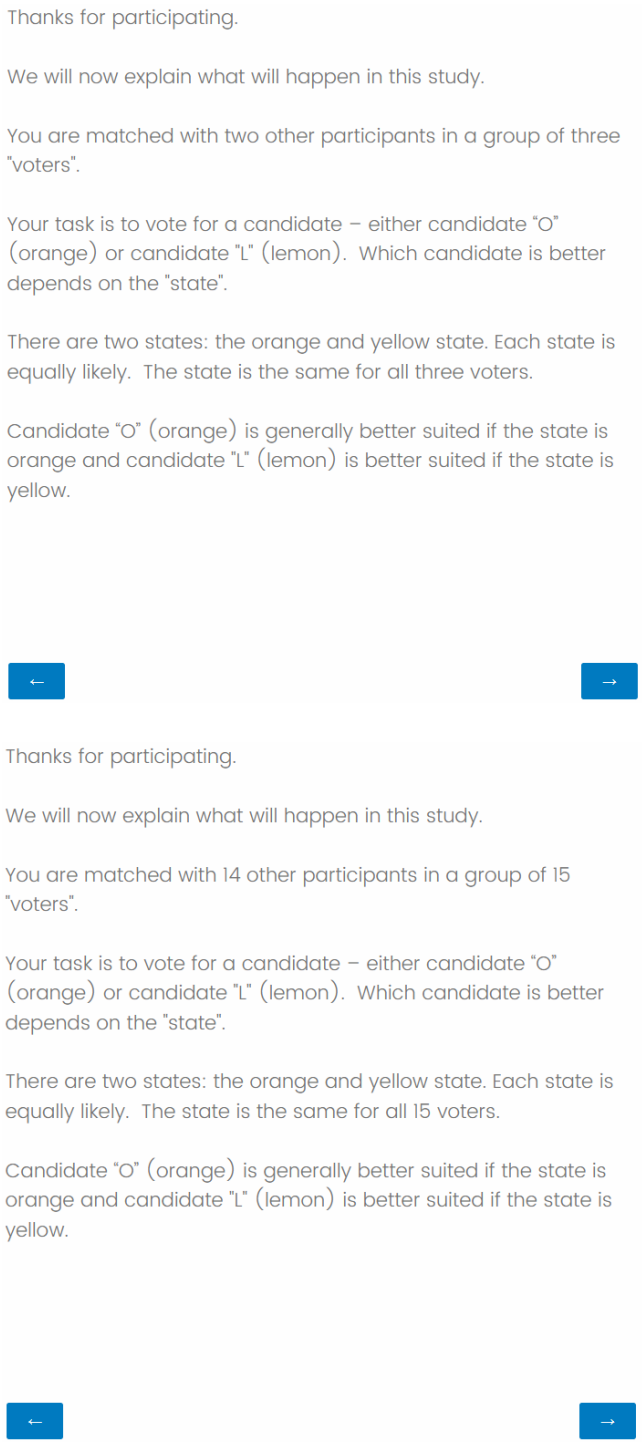


Figure C.2: Prolific ID

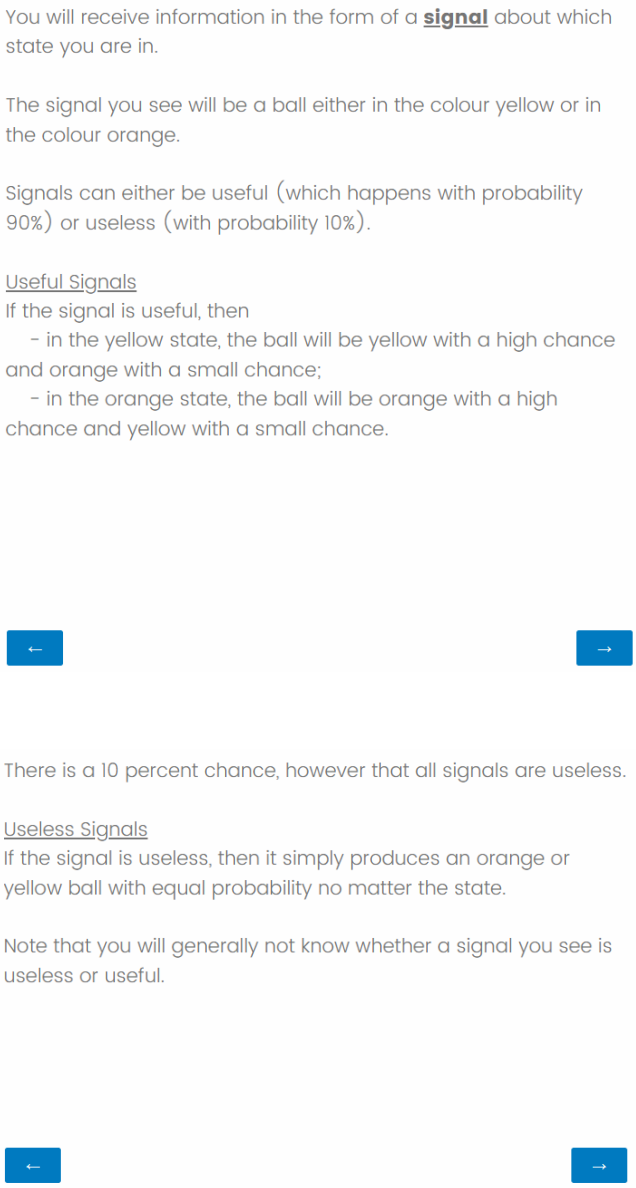


Figure C.3: Instruction – Part 1: Basics of the Experiment.



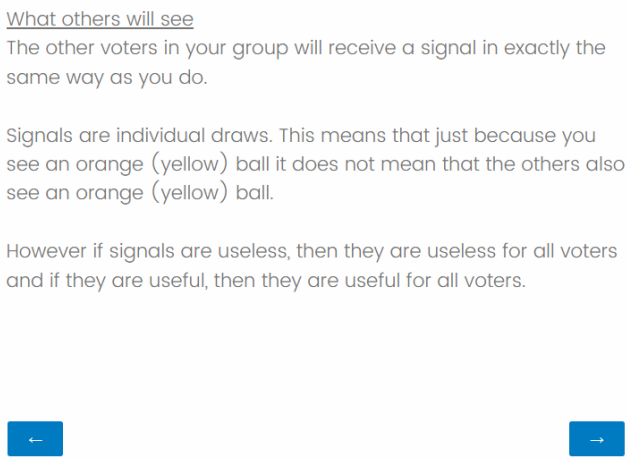
Notes: Participants are informed of the group sizes, the tasks, and the basic information about the experiment.

Figure C.4: Instruction – Part 2: Useful and Useless Signals.



Notes: The probability of signals being useful is fixed throughout the whole experiment and across all treatments.

Figure C.5: Instruction – Part 3: Correlation of Signals



Notes: Participants are informed about the correlation of useless signals.

Figure C.6: Instruction – Part 4: Personal Bias

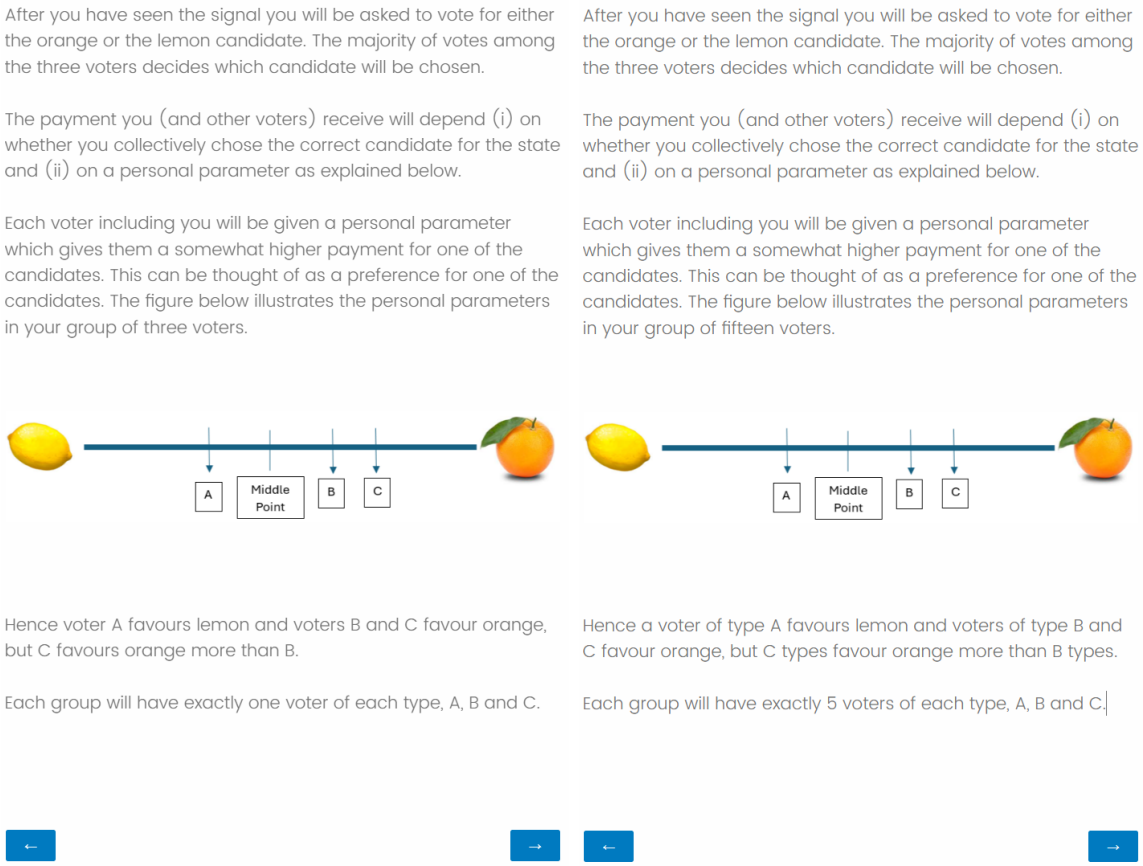


Figure C.7: Instruction – Part 5: Payoffs

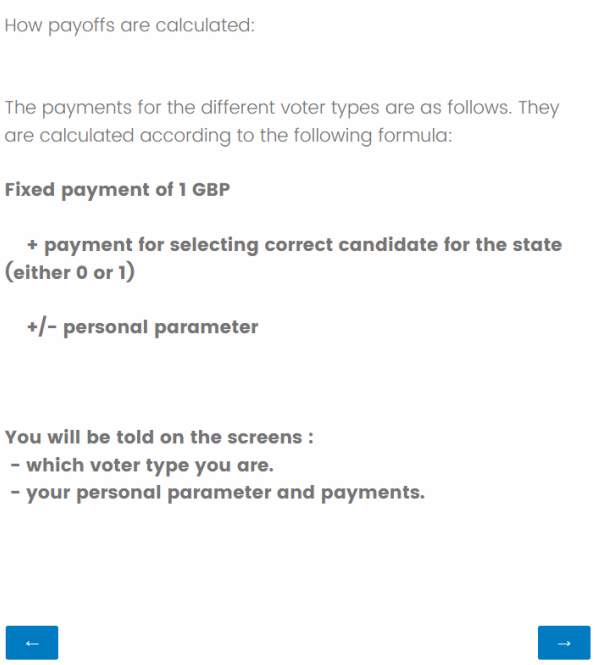


Figure C.8: Instruction – Part 6: More Information about the Experiment

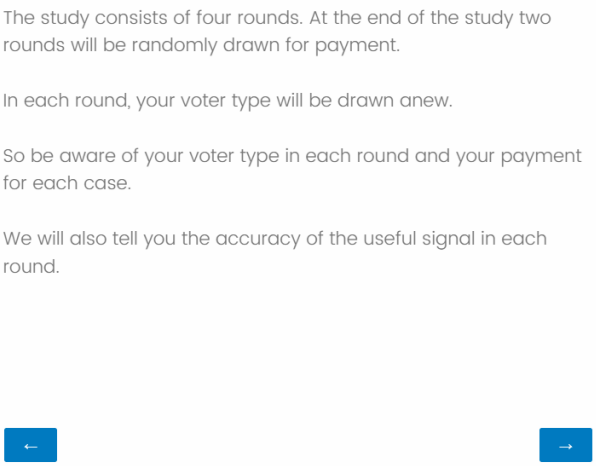


Figure C.9: Start Page Before Each Round

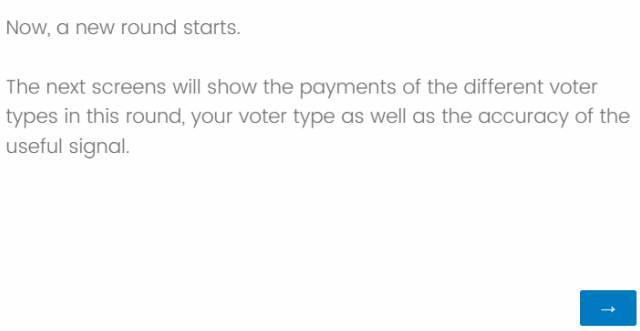
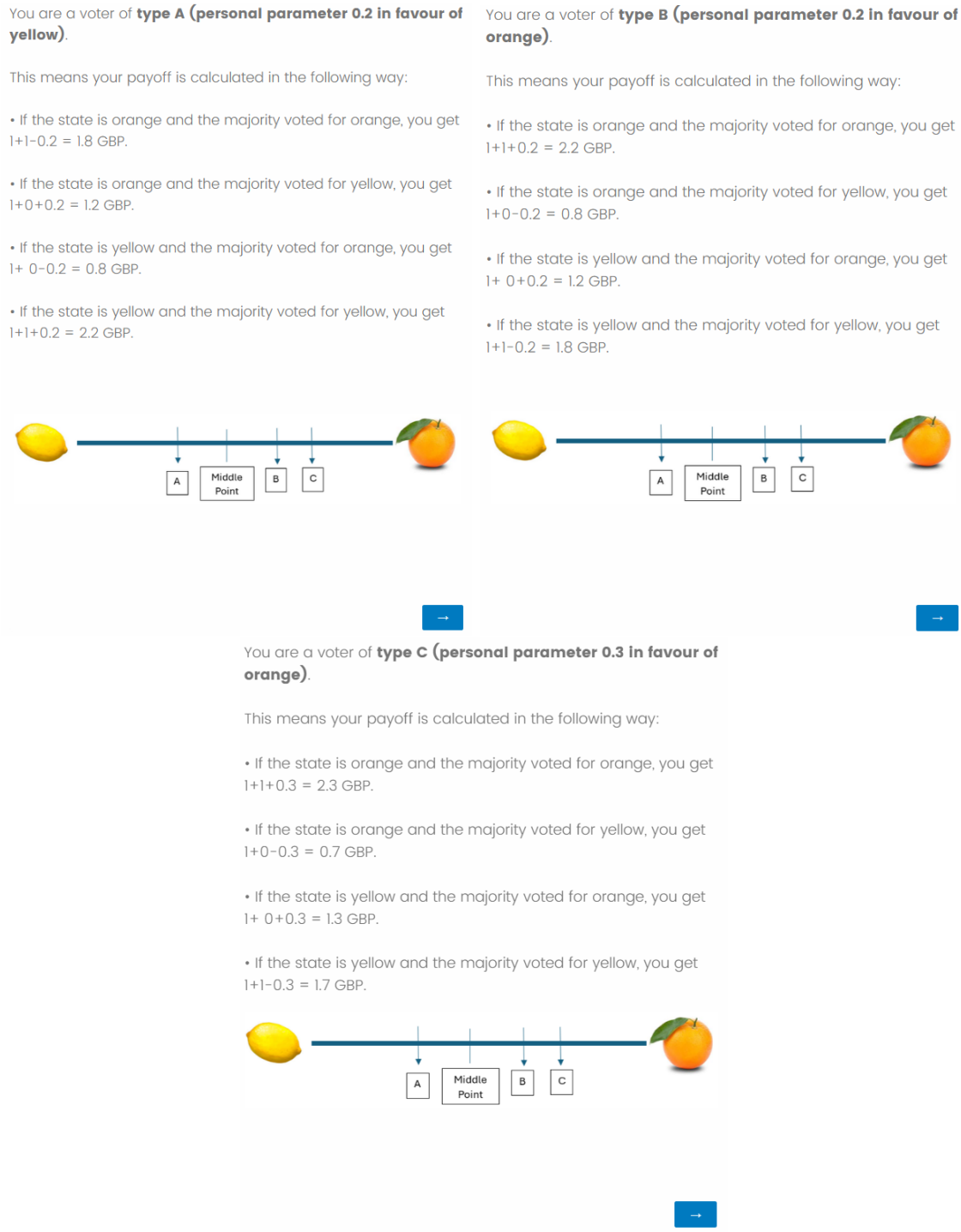


Figure C.10: Details of Payoffs for Each Personal Bias



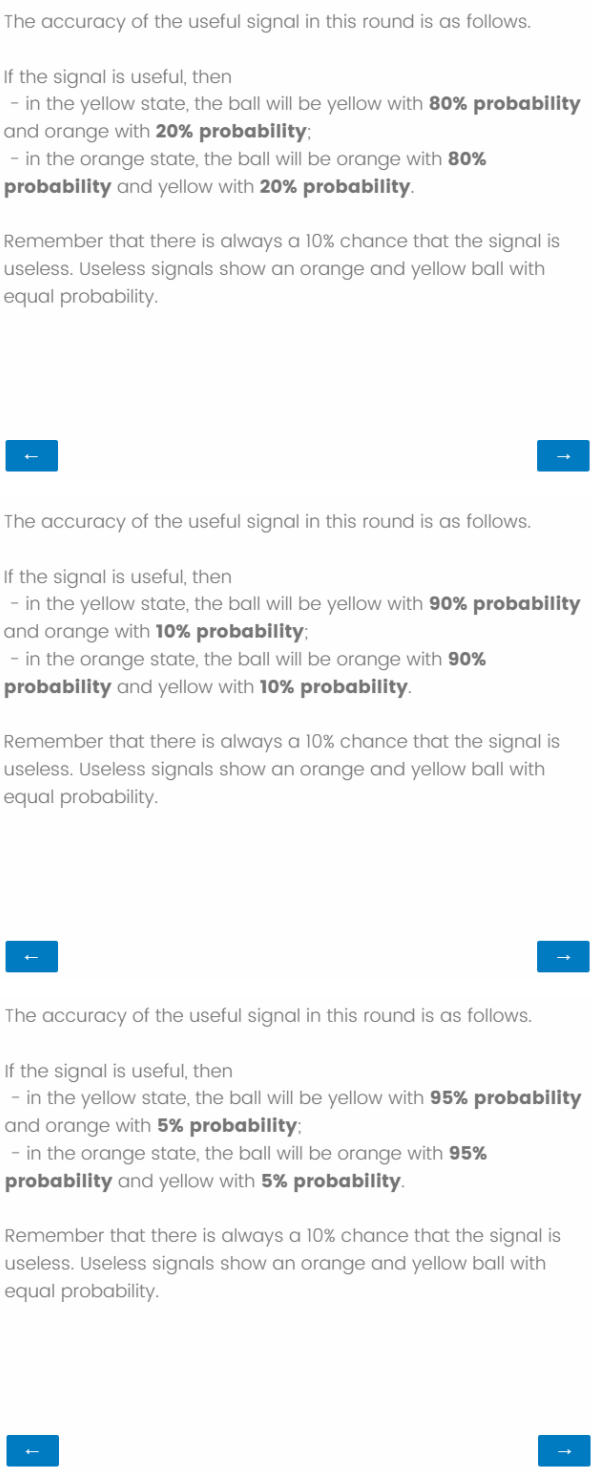
Notes: Participants are informed of the payoff functions under different types and personal biases.

Figure C.11: Reveal the Types of Each Participant and the Corresponding Payoffs for Each Round



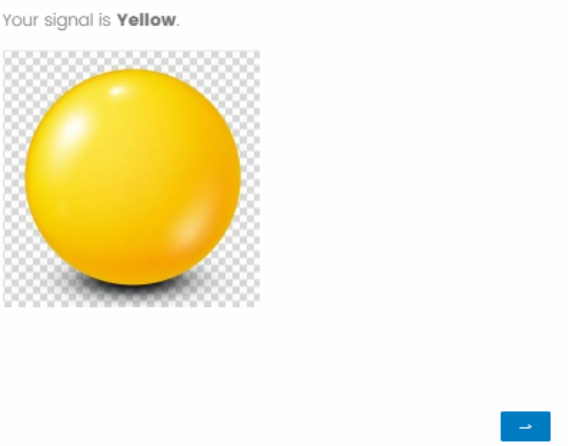
Notes: In each round, one of the three pages is presented to $\frac{1}{3}$ of all participants in the same group. More details about the personal bias are listed in Table 3.

Figure C.12: Reveal the Accuracy of the Useful Signal for Each Round



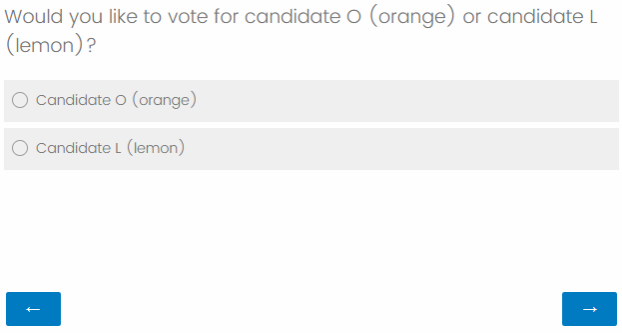
Notes: In each round, one of the three pages is presented to all participants in the same group.

Figure C.13: Reveal the Signal of the True State



Notes: In each round, one of the two balls is presented to each participant. The probability depends on both the accuracy and informativeness of the signal.

Figure C.14: Vote for Candidates



Notes: At the end of each round, participants are required to select one of the two candidates based on all the information they have received above.

Figure C.15: Thanks

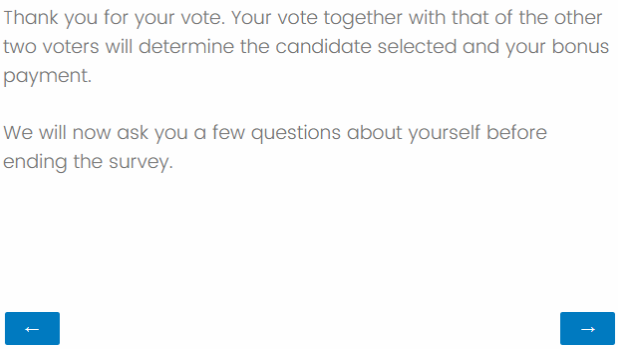


Figure C.16: Questionnaire-1

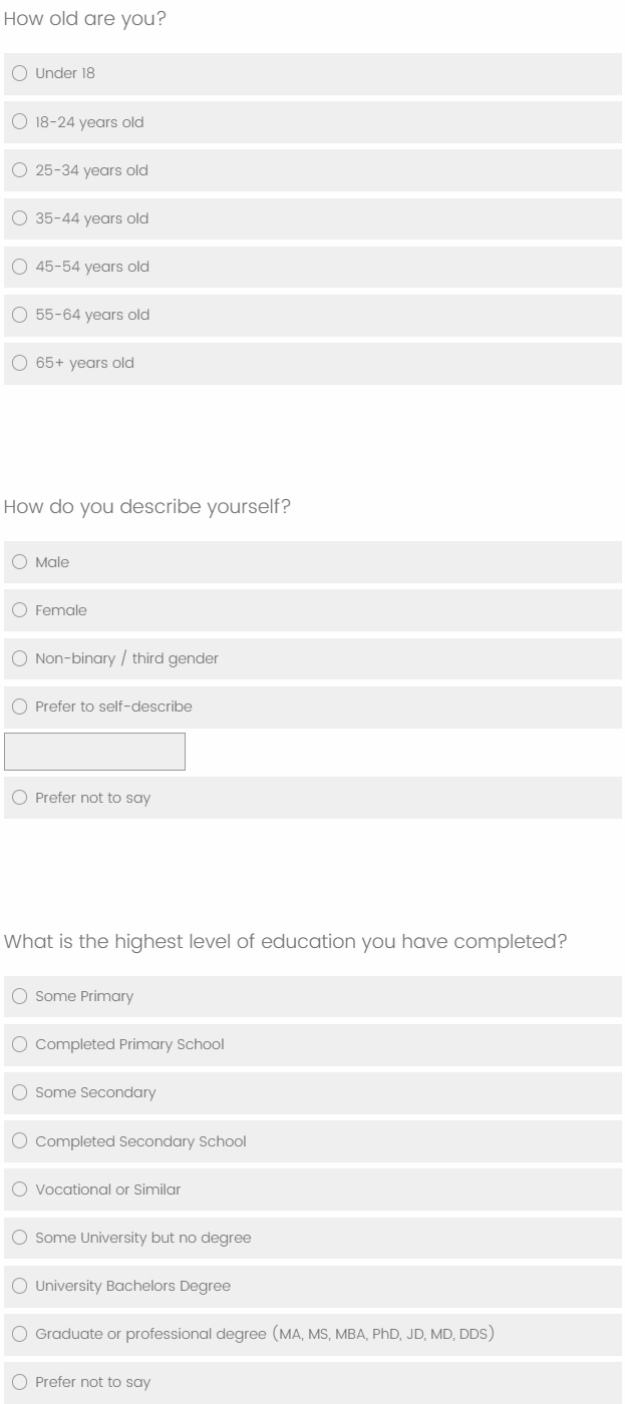


Figure C.17: Questionnaire-2

